

Reviewer Pack — Minimal Rank of Geometric Invariants in Representation Theor...

Entry 69ad7c39854f · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Geometric Invariants in Representation Theory vs Communication Complexity for XOR-AND Networks

Entry ID: 69ad7c39854f **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-22 19:05:04 UTC

1. Conjecture

Field A (mathematical branch): Representation Theory (Geometric Invariants) **Field B** (complexity object): Communication Complexity (XOR-AND Networks)

Statement:

{‘text’: ‘For a given XOR-AND network with n inputs and m outputs, the minimal rank of the geometric invariant associated with the representation theory of its input space is upper bounded by a function $f(m)$ such that $f(n) = O(n^{\{2/3\}})$.’}

Rationale (proposer’s reasoning):

{‘text’: ‘The use of geometric invariants from representation theory could reveal novel structures within XOR-AND networks, potentially leading to communication complexity lower bounds that are not accessible through traditional methods.’}

Taxonomy category: KARCHMER_WIGDERSON (status at proposal time: alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 1a5be04dc9cb692a

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if the mean minimal rank of the geometric invariant across all generated XOR-AND networks satisfies $|\text{mean_rank} - (n^{(2/3)})| \leq 3$ for n inputs, with a p-value less than 0.05 from a statistical test.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): -intitle:geometric invariants AND representation theory AND communication complexity AND XOR-AND networks -title:representation theory AND geometric invariants AND lower bound AND communication complexity -title:XOR-AND networks AND rank AND geometric invariant AND representation theory

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    n = random.randint(5, 40)
    m = random.randint(1, min(n, 10))

    # Generate a random XOR-AND network with n inputs and m outputs
    network = [[random.choice([0, 1]) for _ in range(m)] for _ in range(n)]

    # Compute the minimal rank of the geometric invariant associated with the representation theory
    # This is a placeholder function. Replace it with actual computation if possible.
    min_rank = sum(1 for row in network if any(row))

    return {
        "metric_name": "Minimal Rank",
        "metric_value": min_rank,
        "instances_tested": 1,
        "conjecture_holds": False,
        "counterexample": "mapping_undefined"
    }

if __name__ == "__main__":
    import sys
    seeds = [int(x) for x in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3

    results = []

    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
```

```

results.append(result)

mean_rank = sum(res["metric_value"] for res in results) / len(results)
support_fraction = sum(1 for res in results if res["conjecture_holds"]) / len(results)

RESULT = "SUPPORTED" if support_fraction >= 0.8 else "FALSIFIED"
counterexample = next((res["counterexample"] for res in results if not res["conjecture_holds"]), None)
first_failing_seed = next((res["seed"] for res in results if not res["conjecture_holds"]), None)

print(f"RESULT: {RESULT} mean={mean_rank:.2f} std=0.00 support_fraction={support_fraction:.2f}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

'conjecture_holds': False, 'counterexample': 'mapping_undefined'}
TRIAL: {'metric_name': 'Minimal Rank', 'metric_value': 5, 'instances_tested': 1, 'conjecture_holds':
TRIAL: {'metric_name': 'Minimal Rank', 'metric_value': 25, 'instances_tested': 1, 'conjecture_holds':
TRIAL: {'metric_name': 'Minimal Rank', 'metric_value': 23, 'instances_tested': 1, 'conjecture_holds':
TRIAL: {'metric_name': 'Minimal Rank', 'metric_value': 8, 'instances_tested': 1, 'conjecture_holds':
TRIAL: {'metric_name': 'Minimal Rank', 'metric_value': 30, 'instances_tested': 1, 'conjecture_holds':
TRIAL: {'metric_name': 'Minimal Rank', 'metric_value': 16, 'instances_tested': 1, 'conjecture_holds':
TRIAL: {'metric_name': 'Minimal Rank', 'metric_value': 36, 'instances_tested': 1, 'conjecture_holds':
TRIAL: {'metric_name': 'Minimal Rank', 'metric_value': 19, 'instances_tested': 1, 'conjecture_holds':
TRIAL: {'metric_name': 'Minimal Rank', 'metric_value': 17, 'instances_tested': 1, 'conjecture_holds':

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c466e771.py", line 55, in <module>
    first_failing_seed = next((res["seed"] for res in results if not res["conjecture_holds"]), None)
    ~~~~~^~~~~~
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c466e771.py", line 55, in <genexpr>
    first_failing_seed = next((res["seed"] for res in results if not res["conjecture_holds"]), None)
    ~~~~~^~~~~~
KeyError: 'seed'

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed before producing data, which prevents a definitive evaluation of the conjecture. | next: Investigate and fix the crash in the test code to proceed with the verification of the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	13402
2	preregistration	ollama_remote	glm4:latest	0	0	10121
3	novelty	ollama_remote	glm4:latest	0	0	10192
4	novelty	ollama_remote	glm4:latest	0	0	8748
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14743
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10865
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11336
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	5418
9	judge	ollama_remote	glm4:latest	0	0	12345

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 97168 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/69ad7c39854f.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/69ad7c39854f.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/69ad7c39854f.tar.gz` (if generated)